University Institute of Engineering

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Experiment-3

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Semester: 5th
Subject Name: ADBMS

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Section/Group: 24AIT_KRG-G2
Date of Performance: 27/07/2026
Subject Code:24CSP-333

1. **Aim of the practical:** To perform advanced SQL queries using aggregate functions, joins, subqueries, and ranking techniques for analyzing employee, department, and sales data.

2. **Questions**

(A) Write You are given with the Employee schema; the task is to find the highest score for each respective department.
For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28. Same with the other departments i.e., D2, D3 and D4.

```sql
-----A-----

CREATE TABLE Employee
(
    EID INT PRIMARY KEY,
    DEPT VARCHAR(10),
    SCORES DECIMAL(5,2)
);

INSERT INTO Employee (EID, DEPT, SCORES)
VALUES
(1, 'D1', 1.00),
(2, 'D1', 5.28),
(3, 'D1', 4.00),
(4, 'D2', 8.00),
(5, 'D1', 2.50),
(6, 'D2', 7.00),
(7, 'D3', 9.00),
(8, 'D4', 10.20);
```

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

Solution:

eid [PK] integer	dept character varying (10)	scores numeric (5,2)
1	D1	5.28
2	D1	5.28
3	D1	5.28
4	D2	8.00
5	D1	5.28
6	D2	8.00
7	D3	9.00
8	D4	10.20

(B) A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

```sql
CREATE TABLE SalesPerson
(
    seller_id INT PRIMARY KEY,
    name VARCHAR(50),
    city VARCHAR(50),
    commission INT
);

CREATE TABLE Company
(
    com_id INT PRIMARY KEY,
    name VARCHAR(50),
    city VARCHAR(50)
);

CREATE TABLE Orders
(
    order_id INT PRIMARY KEY,
    order_date DATE,
    com_id INT,
    seller_id INT,
    amount INT,
    FOREIGN KEY (com_id)
        REFERENCES Company(com_id),
    FOREIGN KEY (seller_id)
        REFERENCES SalesPerson(seller_id)
);
```

```sql
INSERT INTO SalesPerson
VALUES
(1,'John','New York',15),
(2,'Amy','Los Angeles',13),
(3,'Mark','Chicago',12),
(4,'Pam','Boston',15);

INSERT INTO Company
VALUES
(1,'RED','Boston'),
(2,'ORANGE','New York'),
(3,'YELLOW','Boston'),
(4,'GREEN','Austin');

INSERT INTO Orders
VALUES
(1,'2024-01-10',1,1,1200),
(2,'2024-01-12',2,1,800),
(3,'2024-01-15',3,2,2500),
(4,'2024-01-18',1,3,1500),
(5,'2024-01-22',4,2,700),
(6,'2024-01-25',2,3,2000),
(7,'2024-01-28',3,4,3000),
(8,'2024-01-30',4,4,200);
```

Solution:

```sql
select seller_id from Orders
group by seller_id
having sum(amount) =
(
select max(max_amt) from
(
select seller_id , SUM(amount) as max_amt
from Orders  group by seller_id
order by seller_id )) ;
```

seller_id integer
3

(C) A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

```
-----C-----
CREATE TABLE Department
(
    id INT PRIMARY KEY,
    name VARCHAR(50)
);

INSERT INTO Department (id, name)
VALUES
(1, 'IT'),
(2, 'Sales');


CREATE TABLE Employee1
(
    id INT PRIMARY KEY,
    name VARCHAR(50),
    salary INT,
    departmentId INT,
    FOREIGN KEY (departmentId)
    REFERENCES Department(id)
);
```

```
INSERT INTO Employee1 (id, name, salary, departmentId)
VALUES
(1, 'Joe',   85000, 1),
(2, 'Henry', 80000, 2),
(3, 'Sam',   60000, 2),
(4, 'Max',   90000, 1),
(5, 'Janet', 69000, 1),
(6, 'Randy', 85000, 1),
(7, 'Will',  70000, 1);
```

Solution:

```
select
    d.name as DEPARTMENT,
    e.name as EMPLOYEE,
    e.salary
from Employee1 as e
join Department as d
on e.departmentId = d.id
where (
    select count (distinct e2.salary)
    from Employee1 as e2
    where
    e2.departmentId = e.departmentId
    and
    e2.salary>e.salary
)<3 order by d.id;
```

department character var	employee character var	salary integer
IT	Joe	85000
IT	Max	90000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

3. Learning Outcomes:

- Understand the use of aggregate functions (SUM(), MAX(), COUNT()) for data analysis.
- Apply GROUP BY and HAVING clauses to generate summarized reports.
- Write subqueries and joins to solve complex SQL problems.
- Analyze employee and sales data to extract meaningful business insights using SQL.